Team Number: 44

Project Title: Breast Cancer Stage Classification with miRNA

Report Date: 10/06/25

Part one: Progress status as a team

All team members have detailed tasks listed on Teams Planner: YES or NO; if NO, explain.

Yes

Please answer the following questions:

1. Have you met as a team this past week? If yes, give date/time, and the members attended the meeting. If no, explain.

   We did not hold formal team meetings; however, we coordinated daily through our team's Discord server to discuss progress, assign tasks, and resolve questions or issues.

2. Have you met with the sponsor as a team? If yes, give date/time, and the member attended the meeting. If no, explain.

   Yes. We met with Dr. Ali Ibrahim on Thursday, October 16, at 6 pm. Joshua Adegboyo, Anika Baro Villa and Brehon Parker were present.

3. Describe verbally the tasks completed the past week and the challenges faced.

   Last week (Week 8) we completed the work started in Week 7 on analyzing the overlap and consistency of selected miRNAs across the different feature selection approaches (traditional, deep learning, and clustering) applied to our breast_cancer dataset. Since the initial feature overlap among deep learning methods was very limited, we increased the number of selected features to k = 15 and k = 20 to improve the likelihood of identifying meaningful overlaps across models (FNN, DAE, 1D-CNN, and LSTM-Seq). Joshua worked on the LSTM model, Anika on the FNN, Nicholas on the CNN, and Brehon on the DAE. After each member generated their top-k miRNA lists, we compared the results and created consensus lists of overlapping miRNAs for both k = 15 and k = 20. Despite increasing k, the overlap across deep learning methods remained limited, highlighting their differing feature selection behaviors. In parallel, we also generated a consensus list of miRNAs across the traditional methods (Chi2, LASSO, SFS, and VT) to complete this stage of feature analysis and prepare for the upcoming classification tasks.

   This week (Week 9) we focused on performing classification using the sets of selected miRNAs from each feature selection approach. Joshua was working with the K-Nearest Neighbors (KNN) classifier, Anika was in charge of implementing the Random Forest model, Nicholas was responsible for the Support Vector Machine (SVM) classifier, and Brehon was working on Logistic Regression. The goal was to evaluate each model's performance using metrics such as accuracy, precision, recall, and macro F1-score to enable a fair comparison across methods. With this analysis we aim to determine which feature selection approach offers the best predictive performance for our dataset and to understand how each method generalizes to the breast cancer classification task.

4. Describe the tasks to be completed the coming week.

   Our team will run stage-specific feature selection for breast cancer stages I through IV using the sponsor's stage-wise setup (Normal vs. Stage X, one run per stage). To do so, we will re-execute our 3 feature selection approaches (traditional, deep learning, and clustering) under the same protocol (same data split, same scaling, same stage framing, and the same cutoffs). For traditional and deep learning methods, we will output the top 5, top 10, and top 15 miRNAs for each stage. For clustering, we will form 2 clusters per stage and then pick the "important" cluster using the sponsor's rule (either literature-based or the smaller-cluster), as directed. Each teammate owns their assigned methods end-to-end, following this shared protocol and delivering per-stage outputs. After everyone finishes their work, we will then aggregate results across methods to build overlap/consensus by stage and move into interpretation and literature validation aligned with our sponsor's feedback.

Previous Report Summary. Date: 9/29/25

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
Joshua Adegboyo	2	0	2
Anika Baro Villa	2	0	2
Nicholas Campos	2	0	2
Brehon Parker	2	0	2

Current Report Summary. Date: 10/06/25

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
Joshua Adegboyo	2	0	3
Anika Baro Villa	5	0	3
Nicholas Campos	2	0	3
Brehon Parker	2	0	3

Name of the Member: Joshua Adegboyo

Report Date: 10/06/25

List the following:

Task 7.1: Continue Neural Network feature selection tests (10/05/25)

In the next round of assignments I will contribute by helping extend the deep learning based feature selection using feedforward neural networks. My focus will be on testing small changes in architecture and tuning hyperparameters such as layer size and activation functions, and evaluating whether the selected miRNAs overlap with those found by unsupervised clustering methods like GMM.

Task 7.2: Prepare project first demo slides (10/05/25)

I will also assist with preparing our demo slides. This will include documenting the results of my Gaussian clustering work, showing how the miRNAs were partitioned, and summarizing our team's progress across classical, clustering, and neural network based feature selection approaches. The goal will be to provide a clear snapshot of both the workflow completed so far and the path forward for the project.

Task 8.1: **Prepare LSTM Feature Outputs for Deep Learning Overlap Analysis (10/13/25)**

This week, I contributed to the deep learning feature selection phase by finalizing my LSTM-Seq based miRNA feature extraction. My original LSTM model produced a list of top 10 significant miRNAs but based on group feedback and Dr. Ali's updated direction to improve feature overlap across deep learning methods, I expanded my results by increasing k to 15 and 20.

These lists were shared with the team and used in the deep learning consensus overlap alongside features from FNN, CNN, and DAE. This step ensured my contribution was included in the final overlap comparison required before classification.

Task 8.2: Generate Final Deep Learning Feature Overlap List (10/14/25)

After preparing my LSTM feature lists, I verified overlap of my LSTM-selected miRNAs with other deep learning methods. The overlap list contained a small number of shared features, which confirmed what was discussed in the team meetings—that deep learning-based feature selection methods capture diverse signals from the data. My updated LSTM outputs (k=15, k=20) were submitted to support the final deep learning overlap file.

Task 9.1: Apply KNN Classification on Selected Feature Groups (10/18/25)

Following Anika's instructions, I applied **K-Nearest Neighbors (KNN)** classification using the **70/30 train-test split** across all three required feature groups:

Feature Group	Input File Used
Traditional consensus features	consensus_traditional_atleast_2.csv
Deep learning consensus features	consensus_dl_atleast_2.csv
Clustering consensus features	consensus_clustering_atleast_2.csv

I also included my LSTM k15 and LSTM k20 feature sets for comparison. StandardScaler was applied before KNN training to normalize feature values. I used k = 5 neighbors, consistent with standard KNN baseline testing.

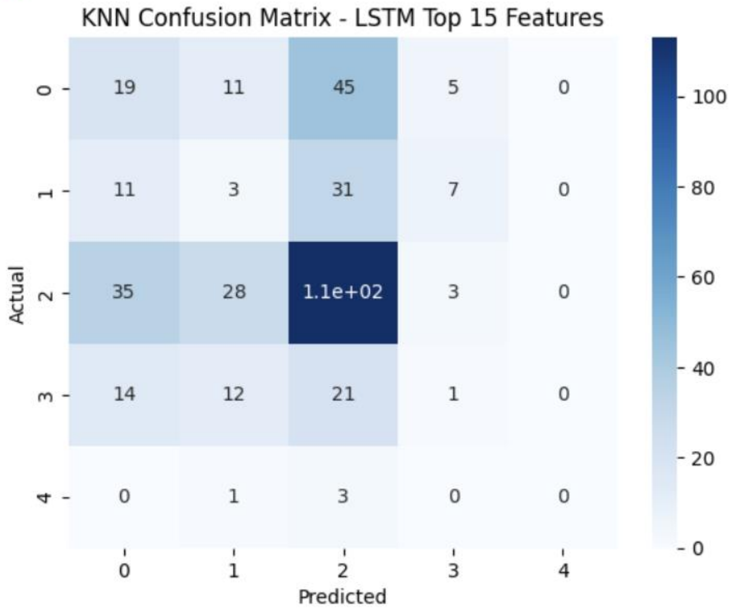
Results Summary (KNN Accuracy):

Feature Set	Accuracy (KNN)
LSTM k=15	~0.37–0.39
LSTM k=20	~0.39–0.40
Traditional Consensus	~0.36
Deep Learning Consensus	~0.35
Clustering Consensus	**~0.38 (Best)**

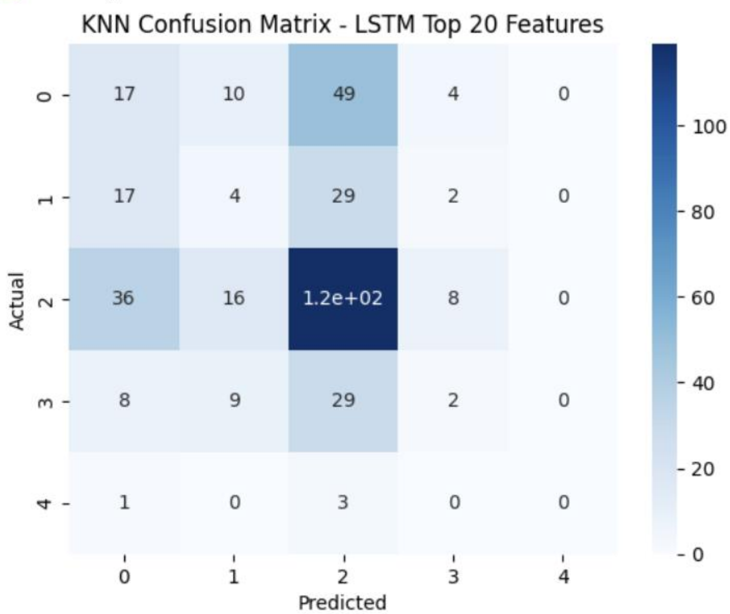
Confusion matrices were also generated for each trial to analyze misclassification patterns across breast cancer stages. As expected, Stage 4 showed the lowest classification performance due to class imbalance.

```
Features loaded:
LSTM k15: 15 features
LSTM k20: 20 features
Traditional: 14 features
Deep Learning: 3 features
Clustering: 116 features
```

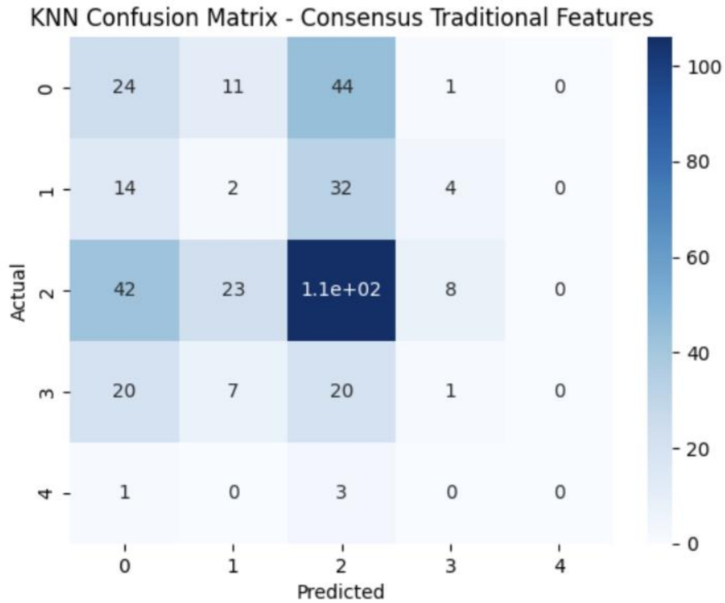
🔷 Running KNN for: LSTM Top 15 Features
✅ Accuracy: 0.3747



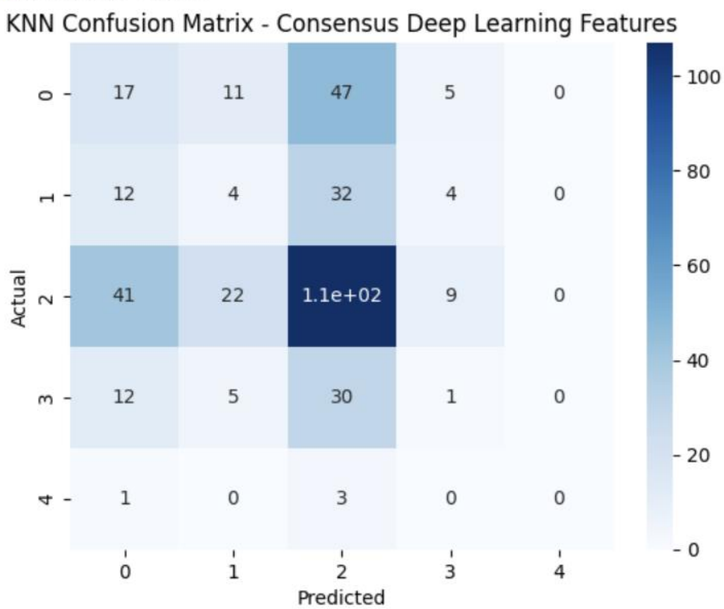
🔷 Running KNN for: LSTM Top 20 Features
✅ Accuracy: 0.3912



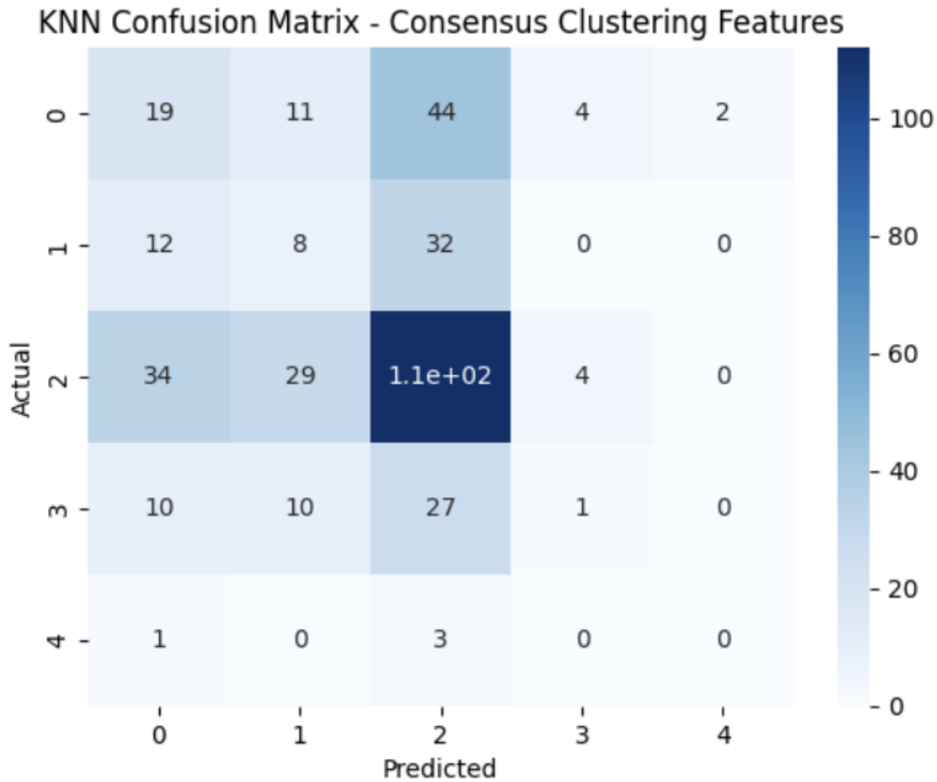
🔷 Running KNN for: Consensus Traditional Features
✅ Accuracy: 0.3664



🔷 Running KNN for: Consensus Deep Learning Features
✅ Accuracy: 0.3554



```
🔷 Running KNN for: Consensus Clustering Features
✅ Accuracy: 0.3857
```



Task 10.1: Apply stage-wise traditional feature selection using LASSO (10/25/25)

For this task, I will extend my previous LASSO feature selection approach to a stage-wise setup to identify stage-specific miRNA biomarkers. The analysis will be conducted separately for each comparison group: Normal vs. Stage I, Normal vs. Stage II, Normal vs. Stage III, and Normal vs. Stage IV. I will use the same preprocessing procedure as before (StandardScaler and 70/30 train-test split) to ensure consistency with prior experiments. Logistic Regression with L1 regularization (LASSO) and class_weight='balanced' will be applied to rank miRNAs by absolute coefficient magnitude. For each stage, I will extract the Top-15 and Top-20 features based on feature importance and export the results as CSV files for cross-method comparison and classification. Model performance (validation accuracy) will be recorded only to verify correct execution and not for final analysis.

Task 10.2: Apply stage-wise deep learning feature selection using LSTM-Seq (10/25/25)

For this task, I will adapt my LSTM sequential model to perform stage-wise deep learning feature selection across the same four stage comparisons used in Task 10.1. For each stage pair (Normal vs. Stage I–IV), I will train a compact bidirectional LSTM architecture with dropout and early stopping to prevent overfitting. Feature importance will be computed using permutation importance over the input dimension, ranking all 1,881 miRNAs based on their contribution to model performance. For each stage model, I will extract the Top-15 and Top-20 most important features and save the results as CSV files. Validation accuracy will be used only to confirm stable model behavior across runs.

Task 10.3: Apply stage-wise clustering as feature selection using K-Means (10/25/25)

In this task, I will implement stage-wise K-Means clustering to extract miRNAs that form discriminative expression patterns for each stage of cancer. Clustering will be applied separately for Normal vs. Stage I, Normal vs. Stage II, Normal vs. Stage III, and Normal vs. Stage IV. Using scaled and transposed data so that miRNA features are clustered instead of samples, I will apply K-Means with k = 2 clusters for each stage. The important cluster will be determined by selecting the cluster with either smaller membership size or stronger variance separation. From this cluster, I will extract the Top-15 miRNAs with the highest variance and save them in stage-specific CSV files. These results will be used later for stage-wise classification analysis.

Name of the Member: Anika Baro Villa

Report Date: 10/06/25

Task 7.1: Generate a list of overlapping miRNAs across different clustering methods (9/29/25)

For this task, I compared the miRNAs within each of the "important" clusters we obtained using different clustering methods (K-Means, Spectral, Gaussian, and Hierarchical) to find overlapping features across them. K-Means's important cluster had 464 miRNAs, Spectral's had 46, Gaussian's just 1 miRNA, while Hierarchical's contained 282 miRNAs. Pairwise comparisons showed limited agreement, with overlaps only between K-Means and Hierarchical (84 miRNAs), Spectral and Hierarchical (31 miRNAs) and GMM and Hierarchical (1 miRNA). I did not find miRNAs that were common to all four methods. These results might indicate that each clustering method captures distinct data structures, with Hierarchical Clustering showing the strongest consistency with other methods.

Task 7.2: Prepare project status report slides (focus on workflow diagram) (10/03/25)

My main focus for this task was creating the workflow diagram for our breast cancer stage classification system. Drawing from our discussions with Dr. Ali Ibrahim, I designed the diagram to clearly show how all major components of the system connect and work together within the overall process. In addition to building the diagram, I revised the entire slide deck, making the necessary adjustments and additions to ensure the presentation was technically accurate and fully aligned with the current progress of our project.

Task 8.1: Generate a list of overlapping miRNAs across different traditional feature selection methods (10/07/25)

For this task, I compared the sets of significant miRNAs identified by each traditional feature selection method we used (Chi2, LASSO, SFS, and VT) to find shared candidates and build a consensus list. The results showed that Chi2 and VT had the highest overlap (8 miRNAs), followed by Chi2 and LASSO (6), LASSO and VT (6), and LASSO and SFS (4), while no miRNAs were common across all four methods. I then computed a frequency-based consensus, finding that 14 miRNAs appeared in at least 2 methods, 5 in at least 3 methods, and none in all 4. I saved the results as separate CSV files, which will be used in the next stages for classification and biological validation.

```python
# Load lists
chi2   = pd.read_csv("Chi2_important_mirnas_simple.csv")['miRNA']
lasso  = pd.read_csv("lasso_significant.csv")['miRNA']
sfs    = pd.read_csv("sfs_significant.csv")['miRNA']
vt     = pd.read_csv("vt_significant.csv")['miRNA']
```

```python
#  Pairwise overlaps (print + save)
pairs = [
    ("Chi2",  chi2_set,  "LASSO", lasso_set),
    ("Chi2",  chi2_set,  "SFS",   sfs_set),
    ("Chi2",  chi2_set,  "VT",    vt_set),
    ("LASSO", lasso_set, "SFS",   sfs_set),
    ("LASSO", lasso_set, "VT",    vt_set),
    ("SFS",   sfs_set,   "VT",    vt_set),
]

for a_name, a_set, b_name, b_set in pairs:
    inter = a_set & b_set
    print(f"\n{a_name} ∩ {b_name}: {len(inter)}")
    if inter:
        sorted_list = sorted(inter)
        print(sorted_list)
        out_fn = f"{a_name.lower()}_{b_name.lower()}_overlap.csv"
        pd.Series(sorted_list, name="miRNA").to_csv(out_fn, index=False)
        print(f"Saved to {out_fn}")
```

```python
# Common to all
common_all = chi2_set & lasso_set & sfs_set & vt_set
print("\nNumber of miRNAs common to ALL methods:", len(common_all))
print("Name of miRNAs common to ALL methods:", list(common_all))
```

```
Chi2 ∩ LASSO: 6
['hsa-let-7b', 'hsa-mir-10a', 'hsa-mir-143', 'hsa-mir-148a', 'hsa-mir-182', 'hsa-mir-30a']
Saved to chi2_lasso_overlap.csv

Chi2 ∩ SFS: 0

Chi2 ∩ VT: 8
['hsa-mir-10a', 'hsa-mir-10b', 'hsa-mir-143', 'hsa-mir-148a', 'hsa-mir-182', 'hsa-mir-21', 'hsa-mir-30a', 'hsa-mir-375']
Saved to chi2_vt_overlap.csv

LASSO ∩ SFS: 4
['hsa-mir-1244-2', 'hsa-mir-3136', 'hsa-mir-4534', 'hsa-mir-644a']
Saved to lasso_sfs_overlap.csv

LASSO ∩ VT: 6
['hsa-mir-10a', 'hsa-mir-143', 'hsa-mir-148a', 'hsa-mir-182', 'hsa-mir-22', 'hsa-mir-30a']
Saved to lasso_vt_overlap.csv

SFS ∩ VT: 0
```

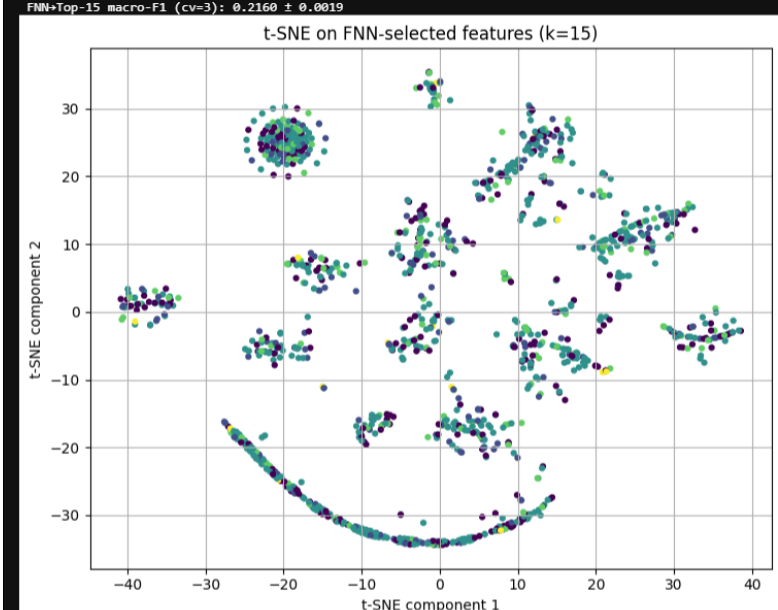
```
Number of miRNAs common to ALL methods: 0
Name of miRNAs common to ALL methods: []
```

consensus_atleast_2	10/7/2025 2:25 PM
consensus_atleast_3	10/7/2025 2:25 PM
consensus_atleast_4	10/7/2025 2:25 PM

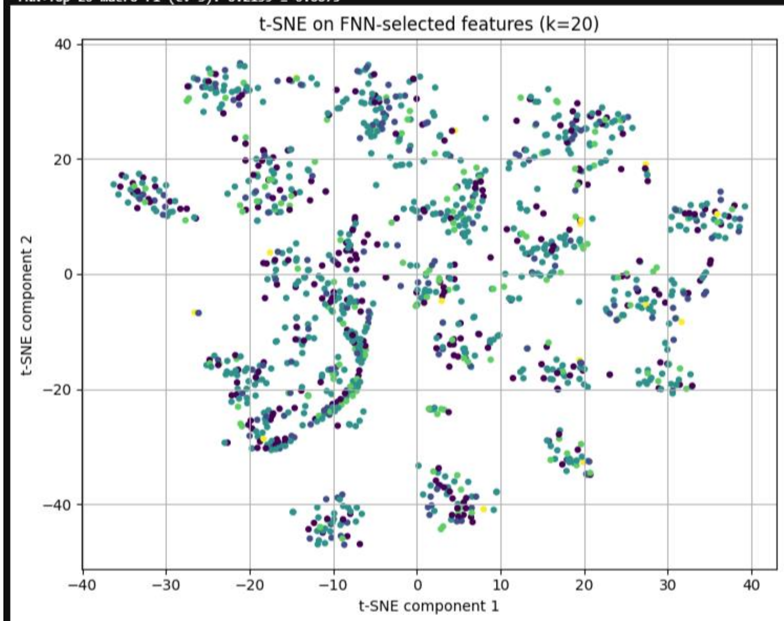
Task 8.2: Expanding FNN feature selection (k = 15, 20) to improve cross-method overlap (10/07/25)

I increased the number of selected miRNAs (k) in the Feed-Forward Neural Network (FNN) I implemented earlier. Previously, I used k values of 3, 5, 8, and 10, so this time I increased k to 15 and 20 to test whether the minimal overlap observed across our deep learning methods was due to overly restrictive cutoff criteria. After training the FNN with L1-regularized input weights and class-balanced loss, I ranked all 1,881 miRNAs by their average absolute weight importance and extracted the top-k subsets. I then evaluated these subsets using a logistic regression classifier with 3-fold cross-validation. I saved the ranked lists and corresponding performance metrics for downstream comparison with the 1D-CNN, AE, and LSTM-Seq results to assess consistency and potential improvements in cross-method overlap.

```
============================================================
FNN Feature Selection (input-L1), k = 15
Selected miRNAs (k=15): ['hsa-mir-2681', 'hsa-mir-3918', 'hsa-mir-549a', 'hsa-mir-4716', 'hsa-mir-4694',
'hsa-mir-3152', 'hsa-mir-4422', 'hsa-mir-4526', 'hsa-mir-1238', 'hsa-mir-548n', 'hsa-mir-1193', 'hsa-mir-5
48aj-1', 'hsa-mir-3621', 'hsa-mir-4705', 'hsa-mir-548b']
FNN→Top-15 macro-F1 (cv=3): 0.2160 ± 0.0019
```



```
================================================================
FNN Feature Selection (input-L1), k = 20
Selected miRNAs (k=20): ['hsa-mir-2681', 'hsa-mir-3918', 'hsa-mir-549a', 'hsa-mir-4716', 'hsa-mir-4694',
'hsa-mir-3152', 'hsa-mir-4422', 'hsa-mir-4526', 'hsa-mir-1238', 'hsa-mir-548n', 'hsa-mir-1193', 'hsa-mir-5
48aj-1', 'hsa-mir-3621', 'hsa-mir-4705', 'hsa-mir-548b', 'hsa-mir-4306', 'hsa-mir-6504', 'hsa-mir-4529',
'hsa-mir-548l', 'hsa-mir-107']
FNN→Top-20 macro-F1 (cv=3): 0.2139 ± 0.0075
```



fnn_k15_features	10/7/2025 11:46 PM
fnn_k20_features	10/7/2025 11:46 PM

Task 8.3: Generate a list of overlapping miRNAs across different deep learning feature selection methods (10/10/25)

For this task, I generated a list of overlapping miRNAs across the deep learning methods (FNN, CNN, LSTM, and DAE) we used for feature selection. I performed the overlap analysis for both k = 15 and k = 20 to evaluate how the number of selected features affected consistency across methods. For k = 15, the overlap was minimal, with only 1 shared miRNA detected, while increasing k to 20 resulted in a slightly higher overlap between methods. No miRNAs were common to all four models in either case. I also computed frequency-based consensus lists, which showed 1 miRNA appearing in at least 2 methods for k = 15 and 3 for k = 20. I exported all results as CSV files for future cross-method and biological comparison.

```python
# Load lists
cnn_list  = pd.read_csv("CNN_important_mirnas_15.csv")['miRNA']
fnn_list  = pd.read_csv("fnn_k15_features.csv")['miRNA']
lstm_list = pd.read_csv("lstm_k15_features.csv")['miRNA']
dae_list  = pd.read_csv("dae_k15_features.csv")['miRNA']
```

```
CNN ∩ FNN: 0
CNN ∩ LSTM: 1
['hsa-mir-1203']
Saved to cnn_lstm_overlap.csv
CNN ∩ DAE: 0
FNN ∩ LSTM: 0
FNN ∩ DAE: 0
LSTM ∩ DAE: 0
```

```
Number common to ALL methods: 0
Names common to ALL methods: []
```

```
# Load lists
cnn_list  = pd.read_csv("CNN_important_mirnas_20.csv")['miRNA']
fnn_list  = pd.read_csv("fnn_k20_features.csv")['miRNA']
lstm_list = pd.read_csv("lstm_k20_features.csv")['miRNA']
dae_list  = pd.read_csv("dae_k20_features.csv")['miRNA']
```

```
CNN ∩ FNN: 0
CNN ∩ LSTM: 2
['hsa-mir-1203', 'hsa-mir-1469']
Saved to cnn_lstm_overlap.csv
CNN ∩ DAE: 0
FNN ∩ LSTM: 0
FNN ∩ DAE: 0
LSTM ∩ DAE: 1
['hsa-mir-136']
Saved to lstm_dae_overlap.csv
```

```
Number common to ALL methods: 0
Names common to ALL methods: []
```

```
consensus_dl_k15_atleast_2    10/10/2025 10:39 PM
consensus_dl_k20_atleast_2    10/10/2025 10:39 PM
```

Task 9.1: Apply Random Forest classification on miRNAs selected by feature selection approaches (10/15/25)

I applied Random Forest classification to the sets of miRNAs that were selected using our 3 main feature selection approaches: traditional methods, deep learning, and clustering. For each approach, I used the same preprocessing and evaluation pipeline to maintain consistency across all experiments. This pipeline included applying a log1p transformation to the data and training a Random Forest classifier using cross-validation. I used accuracy as the main evaluation metric, as required by Dr. Ali Ibrahim, and recorded additional metrics such as macro-F1 and per-class recall to better understand how the model performed across all breast cancer stages. To ensure the results were reliable and not affected by data leakage, all transformations and model training were done within each cross-validation fold. After completing the classification for each of the 3 feature selection approaches, I summarized the mean and standard deviation of the cross-validated accuracy values to compare the model's performance when using miRNAs selected by different methods. The following screenshots show the Random Forest classification results for the consensus feature sets obtained from the traditional, deep learning, and clustering methods.

```
RANDOM FOREST CLASSIFICATION RESULTS
Feature Set: consensus_traditional_atleast_2.csv
Number of Features: 14
Number of Samples: 1209

Best Hyperparameters (via GridSearchCV):
  rf__max_depth: 15
  rf__max_features: sqrt
  rf__n_estimators: 500

Grid Search Best Score (accuracy): 0.4174

Metric                      Score
----------------------------------------
Accuracy [PRIMARY]          0.4123 ± 0.0297
F1-Score (Macro)            0.1831 ± 0.0205
Precision (Macro)           0.1896 ± 0.0341
Recall (Macro)              0.1996 ± 0.0180
F1-Score (Weighted)         0.3539 ± 0.0239

Best accuracy: 0.4123 ± 0.0297
```

```
RANDOM FOREST CLASSIFICATION RESULTS
Feature Set: consensus_dl_atleast_2.csv
Number of Features: 3
Number of Samples: 1209

Best Hyperparameters (via GridSearchCV):
  rf__max_depth: 10
  rf__max_features: sqrt
  rf__n_estimators: 200

Grid Search Best Score (accuracy): 0.2548

Metric                    Score
----------------------------------------
Accuracy [PRIMARY]        0.2535 ± 0.0226
F1-Score (Macro)          0.1871 ± 0.0178
Precision (Macro)         0.2017 ± 0.0157
Recall (Macro)            0.2065 ± 0.0332
F1-Score (Weighted)       0.2762 ± 0.0238

Best accuracy: 0.2535 ± 0.0226
```

```
RANDOM FOREST CLASSIFICATION RESULTS
Feature Set: consensus_clustering_atleast_2.csv
Number of Features: 116
Number of Samples: 1209

Best Hyperparameters (via GridSearchCV):
  rf__max_depth: 15
  rf__max_features: 0.3
  rf__n_estimators: 500

Grid Search Best Score (accuracy): 0.4660

Metric                    Score
----------------------------------------
Accuracy [PRIMARY]        0.4748 ± 0.0134
F1-Score (Macro)          0.1630 ± 0.0126
Precision (Macro)         0.2179 ± 0.0574
Recall (Macro)            0.2055 ± 0.0084
F1-Score (Weighted)       0.3507 ± 0.0132

Best accuracy: 0.4748 ± 0.0134
```

Task 9.2: Compare classification performance between traditional, deep learning, and clustering methods (10/16/25)

I compared the classification performance of the Random Forest models I created across the traditional, deep learning, and clustering-based feature selection approaches. The clustering-derived feature set contained 116 miRNAs and achieved the highest cross-validated accuracy of approximately 0.47 with the lowest variability across folds. The traditional feature set included 14 miRNAs and reached an accuracy of about 0.41, showing stable but moderate performance. The deep learning consensus set had only 3 overlapping miRNAs and achieved a lower accuracy of around 0.25, which was expected given the small number of features and the difficulty of the five-class stage classification task. In all cases, accuracy was used as the primary metric for comparison, while macro-F1 and per-class recall were reviewed to understand class imbalance effects. All models showed low recall for stage 4 samples, indicating that additional strategies may be needed to improve performance on minority classes. Overall, this task helped me evaluate how well Random Forest performed across different sets of miRNAs and highlighted that the clustering-derived features provided the strongest predictive signal among the 3 approaches.

Task 10.1: Apply stage-wise traditional feature selection using SFS (10/25/25)

For this task, I will re-run the Sequential Feature Selection (SFS) method separately for each breast cancer stage (Normal vs. Stage I–IV) to generate stage-specific feature lists. The setup will use the same preprocessing pipeline with StandardScaler and Logistic Regression (L2 regularization, class_weight='balanced') to ensure consistency across runs. For each stage, I plan to extract the Top-5, Top-10, and Top-15 most informative miRNAs, saving the results as separate CSV files. I will use performance metrics only as a sanity check to confirm that the feature selection behaved as expected.

Task 10.2: Apply stage-wise deep learning feature selection using FNN (10/25/25)

This task will extend the same stage-wise approach using a small, regularized Feedforward Neural Network (FNN). The model will include both L1 and L2 regularization, dropout, and early stopping to reduce overfitting. For each run (Normal vs. Stage I–IV), I will rank the miRNAs based on the absolute values of the input-to-hidden layer weights averaged over multiple seeds, then extract the Top-5, Top-10, and Top-15 miRNAs per stage. I will report test accuracy and AUC only as a quick check of model reliability.

Task 10.3: Apply stage-wise clustering as feature selection using K-Means (10/25/25)

For this task, I will apply K-Means clustering as a stage-wise feature selection method. I will conduct the analysis separately for each stage comparison (Normal vs. Stage I–IV) using the same preprocessing and data transposition protocol as before. For each stage, I will run K-Means with 2 clusters to identify groups of miRNAs that share similar expression patterns. Based on Dr. Ali's guidance, the "important" cluster will be determined either by literature evidence or by selecting the smaller of the 2 clusters.

Name of the Member: Nicholas Campos

Report Date: 10/06/25

Task 7.1: Cross-Method Feature Comparison and Data Organization (10/4/25)

Building on both my Chi2 statistical feature selection and 1D-CNN deep learning implementations, I systematically organized and saved the important miRNA biomarkers identified by each method for comprehensive cross comparison. I created standardized CSV output files for both Chi2 and CNN approaches, generating detailed versions containing miRNA names with indices and simplified versions containing only miRNA names for easier comparison with other team methods like hierarchical clustering and K-means. The Chi2 method aggregated features selected across k=[10, 8, 3] into a unified important features list, while the CNN approach similarly collected unique features across the same k values. This organizational structure enables direct comparison between traditional statistical methods (Chi2), unsupervised clustering methods (hierarchical clustering), and deep learning approaches (1D-CNN) to identify consensus biomarkers that emerge across multiple independent methodologies. The deliverable includes properly formatted CSV files maintaining consistent naming conventions (Chi2_important_mirnas_list.csv, CNN_important_mirnas_list.csv, and their simple versions) ready for team integration and overlap analysis with other feature selection approaches.

Task 7.2: Prepare Project First Demo Slides (Objective and Approach slide) (10/05/25)

I created the "Objective and Approach" slide summarizing our breast cancer stage classification project for the first demo presentation. The slide is divided into two main sections: Objective and Approach. The Objective section outlines the problem of inaccurate breast cancer staging methods, our goal to compare traditional ML versus deep learning for miRNA-based classification (Stage I-IV), and our innovation of using strict 70/15/15 data isolation to prevent leakage plus comprehensive feature selection comparison. The Approach section details our data source of 1881 miRNAs from TCGA with clinical staging and presents our pipeline components in four visual categories: Traditional Methods (SFS, VT, Chi2, LASSO), Deep Learning (FNN, DAE, 1D-CNN, LSTM-Seq), Clustering (K-Means, SC, GMM, HC), and Classifiers (KNN, Random Forest, LR, SVM). The slide uses a professional dark theme with breast cancer awareness pink accents and clear visual hierarchy for effective presentation delivery.

```
# Save Chi2 important miRNAs to CSV files
import pandas as pd

chi2_important_mirnas_data = []
for idx in sorted(chi2_top_features):
    mirna_name = miRNA_names[idx]
    chi2_important_mirnas_data.append({'miRNA_name': mirna_name, 'index': idx})

# Save Chi2 detailed CSV
chi2_important_df = pd.DataFrame(chi2_important_mirnas_data)
chi2_important_df.to_csv('Chi2_important_mirnas_list.csv', index=False)

# Save Chi2 simple CSV (just miRNA names)
chi2_important_simple = pd.DataFrame({'miRNA': [miRNA_names[idx] for idx in sorted(chi2_top_features)]})
chi2_important_simple.to_csv('Chi2_important_mirnas_simple.csv', index=False)

print(f"Chi2 files saved: {len(chi2_important_mirnas_data)} miRNAs")
print(f"- Chi2_important_mirnas_list.csv")
print(f"- Chi2_important_mirnas_simple.csv")
```

```
Chi2 files saved: 10 miRNAs
- Chi2_important_mirnas_list.csv
- Chi2_important_mirnas_simple.csv
```

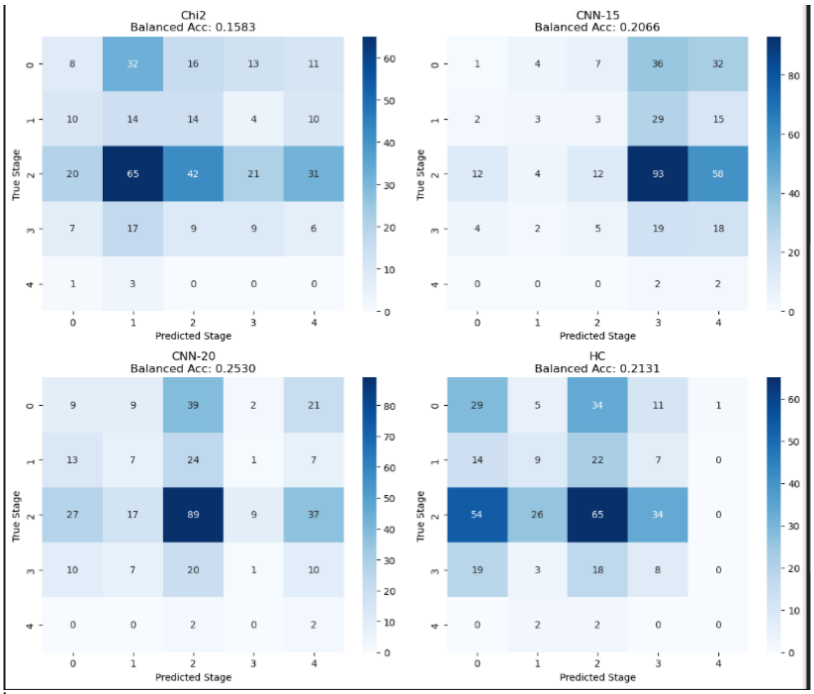
```python
# Save 1D-CNN important miRNAs to CSV files
import pandas as pd

# Collect all unique CNN-selected features across different k values
cnn_all_features = set()
for k in [10, 8, 3]:
    cnn_all_features.update(cnn_results[k]['indices'])

# Create data structures for saving
cnn_important_mirnas_data = []
for idx in sorted(cnn_all_features):
    mirna_name = miRNA_names[idx]
    cnn_important_mirnas_data.append({'miRNA_name': mirna_name, 'index': idx})

# Save CNN detailed CSV
cnn_important_df = pd.DataFrame(cnn_important_mirnas_data)
cnn_important_df.to_csv('CNN_important_mirnas_list.csv', index=False)

# Save CNN simple CSV (just miRNA names)
cnn_important_simple = pd.DataFrame({'miRNA': [miRNA_names[idx] for idx in sorted(cnn_all_features)]})
cnn_important_simple.to_csv('CNN_important_mirnas_simple.csv', index=False)

print(f"CNN files saved: {len(cnn_important_mirnas_data)} miRNAs")
print(f"- CNN_important_mirnas_list.csv")
print(f"- CNN_important_mirnas_simple.csv")
```

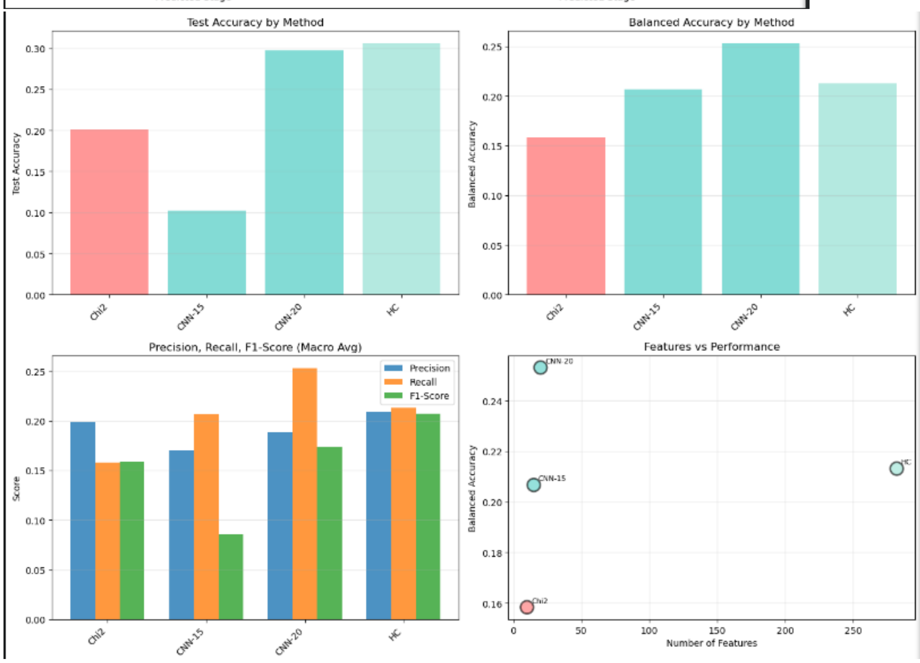
```
CNN files saved: 10 miRNAs
- CNN_important_mirnas_list.csv
- CNN_important_mirnas_simple.csv
```

Task 8.1: SVM Classification on Selected miRNA Features (10/9/25)

Following Dr. Ali's instructions, I implemented Support Vector Machine (SVM) classification to evaluate the miRNA features selected from three distinct feature selection approaches: traditional methods (Chi2), deep learning (1DCNN), and clustering (Hierarchical Clustering). For each approach, I loaded the corresponding important miRNA CSV files, extracted those specific features from the dataset, and trained separate SVM classifiers maintaining the same 70/30 data split protocol for consistency. The SVM models were trained with optimized hyperparameters to predict breast cancer stage classification (Stage I-IV). The classification results provided comprehensive performance metrics including accuracy, precision, recall, and F1-scores for each feature selection method. The deliverable includes documented accuracy metrics for all three approaches, confusion matrices showing classification performance across cancer stages, and trained SVM models ready for comparative analysis with other team members' classifiers.

```
================================================================================
COMPREHENSIVE PERFORMANCE SUMMARY - ALL METHODS
================================================================================

 Method      Category  Features Test Acc Balanced Acc Precision Recall F1-Score
   Chi2   Traditional        10   0.2011       0.1583    0.1982 0.1583   0.1585
 CNN-15 Deep Learning        15   0.1019       0.2066    0.1700 0.2066   0.0855
 CNN-20 Deep Learning        20   0.2975       0.2530    0.1884 0.2530   0.1736
     HC    Clustering       282   0.3058       0.2131    0.2089 0.2131   0.2072

🏆 BEST METHOD: CNN-20
   Balanced Accuracy: 0.2530
   Category: Deep Learning

================================================================================
SAVING RESULTS TO FILES
================================================================================
```

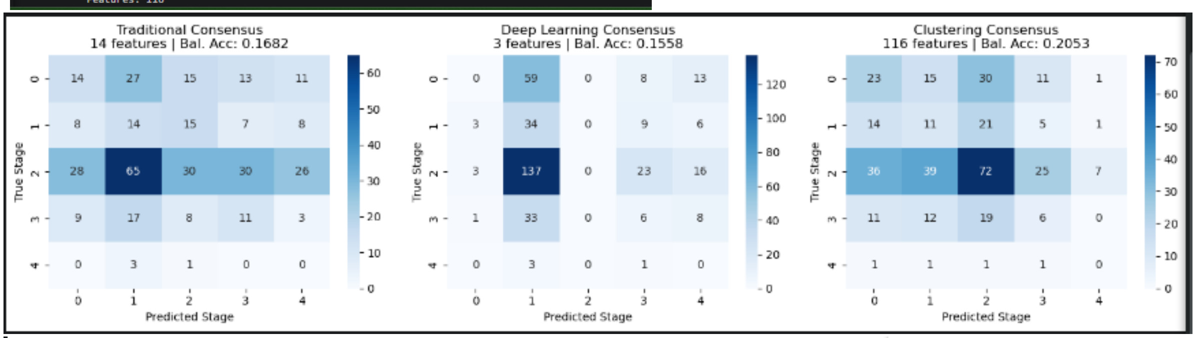

Chi2
Balanced Acc: 0.1583
CNN-15
Balanced Acc: 0.2066
CNN-20
Balanced Acc: 0.2530
HC
Balanced Acc: 0.2131
True Stage
Predicted Stage

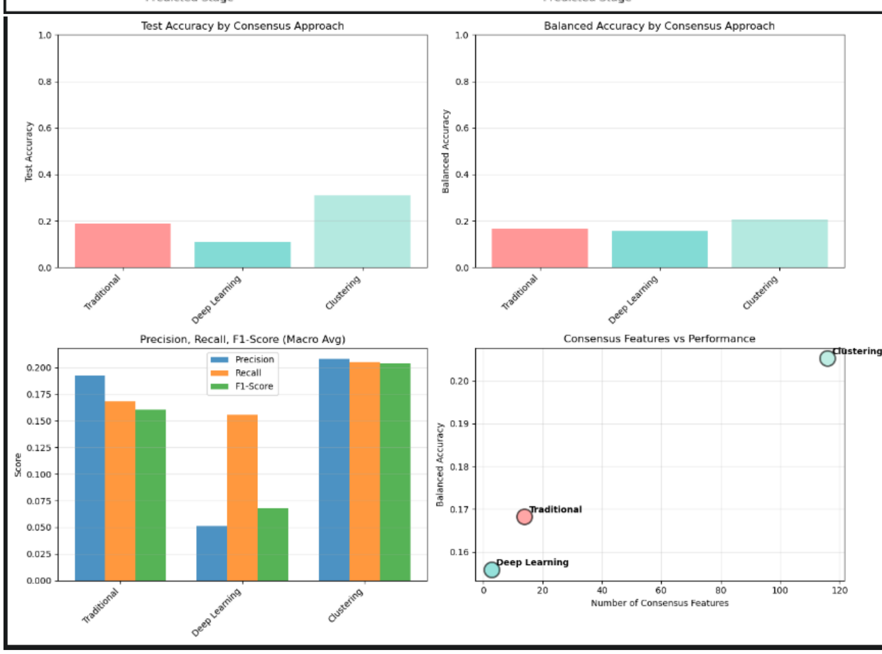

Test Accuracy by Method
Test Accuracy
Balanced Accuracy by Method
Balanced Accuracy
Chi2
CNN-15
CNN-20
HC
Precision, Recall, F1-Score (Macro Avg)
Score
Precision
Recall
F1-Score
Features vs Performance
Number of Features

Task 8.2: Cross-Classifier Performance Comparison (10/9/25)

Building on my SVM classification results, I will collaborate with my team members to conduct a comprehensive comparison of classifier performance across different algorithms as requested by Dr. Ali. Each team member will apply a different classifier (KNN, Random Forest, Logistic Regression, and my SVM) to the same three sets of selected miRNA features from traditional, deep learning, and clustering methods. This systematic comparison will reveal which combination of feature selection method and classifier produces the best results for breast cancer stage classification. I will compile accuracy metrics from all team members' classifiers and create a comparative analysis matrix showing performance differences across methods. The analysis will identify the optimal feature selection approach by examining which method consistently produces high accuracy across multiple classifiers, and determine whether traditional statistical methods, deep learning-based selection, or unsupervised clustering yields superior features for cancer stage prediction. This comparison will directly address Dr. Ali's question about which method gives better results and provides evidence-based recommendations for the final diagnostic system implementation.

```
TRAINING SVM ON ALL CONSENSUS FEATURE SETS
================================================================================

================================================================================
Traditional Consensus
Description: Consensus from Traditional Methods (Chi2, etc.)
Agreement: At least 2 team members | Features: 14
================================================================================

Training Balanced SVM (class_weight='balanced')...

Performance Metrics:
  Training Accuracy:        0.3747
  Test Accuracy:            0.1901
  Test Balanced Accuracy:   0.1682
  Test Precision (macro):   0.1927
  Test Recall (macro):      0.1682
  Test F1-Score (macro):    0.1605

================================================================================
Deep Learning Consensus
Description: Consensus from Deep Learning Methods (CNN, etc.)
Agreement: At least 2 team members | Features: 3
================================================================================

Training Balanced SVM (class_weight='balanced')...

Performance Metrics:
  Training Accuracy:        0.1584
  Test Accuracy:            0.1102
  Test Balanced Accuracy:   0.1558
  Test Precision (macro):   0.0511
  Test Recall (macro):      0.1558
  Test F1-Score (macro):    0.0680

================================================================================
Clustering Consensus
Description: Consensus from Clustering Methods (HC, K-means, etc.)
Agreement: At least 2 team members | Features: 116
================================================================================

Training Balanced SVM (class_weight='balanced')...

Performance Metrics:
  Training Accuracy:        0.5887
  Test Accuracy:            0.3085
  Test Balanced Accuracy:   0.2053
  Test Precision (macro):   0.2080
  Test Recall (macro):      0.2053
  Test F1-Score (macro):    0.2040

================================================================================
COMPREHENSIVE PERFORMANCE SUMMARY - CONSENSUS FEATURES
================================================================================

      Approach              Agreement  Features Test Acc Balanced Acc Precision Recall F1-Score
  Traditional At least 2 team members        14   0.1901       0.1682    0.1927 0.1682   0.1605
Deep Learning At least 2 team members         3   0.1102       0.1558    0.0511 0.1558   0.0680
   Clustering At least 2 team members       116   0.3085       0.2053    0.2080 0.2053   0.2040

🏆 BEST CONSENSUS APPROACH: Clustering
   Balanced Accuracy: 0.2053
   Features: 116
```





10.1: Apply stage-wise traditional feature selection using Chi-Square (10/25/25)

In this task, I'll perform Chi-Square statistical analysis individually for each cancer stage comparison (Normal vs. Stage I, Normal vs. Stage II, etc.) to identify stage-specific important miRNA features. My approach maintains consistency by using StandardScaler for normalization and SVM with balanced class weighting as the classifier. For every stage comparison, I'll identify the Top-15 and Top-20 features with the highest chi-square statistical scores and export these to CSV format. Performance metrics will serve primarily to verify the feature selection process is working correctly.

10.2: Apply stage-wise deep learning feature selection using 1D-CNN (10/25/25)

I'll implement a 1D Convolutional Neural Network with an attention layer to perform stage-by-stage feature selection. The network architecture incorporates dropout layers and early stopping mechanisms to prevent overfitting during training. My workflow involves running separate models for each stage comparison (Normal vs. Stage I–IV), then extracting the Top-15 and Top-20 miRNAs by analyzing the learned convolutional weights and computing feature importance rankings. Accuracy and balanced accuracy metrics will be calculated to ensure the model training was successful.

10.3: Apply stage-wise clustering as feature selection using Hierarchical Clustering (10/25/25)

This task focuses on applying Hierarchical Clustering with Ward linkage to discover important features for each stage independently. I'll run separate clustering analyses comparing Normal samples against each cancer stage (I through IV), following consistent data preprocessing and transposition steps. Rather than using cluster membership alone, I'll apply a variance-based approach to select the most informative miRNAs—specifically, choosing the top 15% of features ranked by their variance across patient samples. Each stage's important features will be documented in individual CSV files.

Name of the Member: Brehon Parker

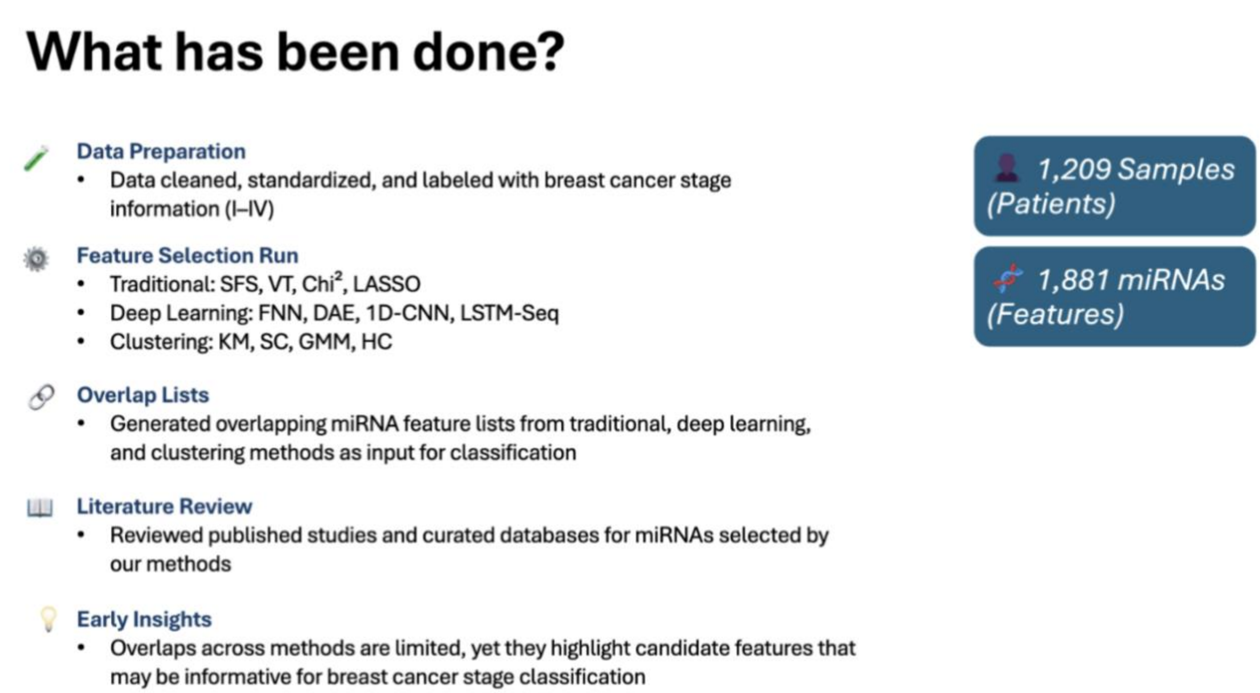
Report Date: 10/06/25

Task 7.1: **Traditional (VT) and Deep Learning (DAE) Feature Outputs for Overlap Analysis (10/04/25)**

Using both my Variance Threshold (traditional) and Denoising Autoencoder (deep learning) feature selection approaches, I prepared and saved the lists of significant miRNAs identified by each method. For Variance Threshold, I exported the selected features into standardized CSV outputs, formatted for consistency with the team. For DAE, I generated top-10 feature lists, ensuring they align with the simplified naming structure for later overlap comparison. These deliverables ensure that my results can be directly integrated with other team outputs (Chi2, CNN, clustering methods, etc.) to create overlap lists across traditional, deep learning, and clustering pipelines.

Task 7.2: Prepare project first demo slides (10/04/25)

For the first demo presentation, I worked on the dataset overview slide. I introduced the scope of our data — 1,209 patient samples and 1,881 miRNAs — to give context before diving into feature selection and classification. At first, I added bar and scatter plots, but after reviewing them with the team, I realized they didn't really add much value, so I removed them. Instead, I focused on keeping the slide clean, consistent with the rest of the deck, and easy to follow. My goal was to make sure the dataset was presented clearly without unnecessary clutter, so the audience can quickly understand the foundation of our work.



Task 8.2: **Apply Logistic Regression Classification on miRNAs Selected by Feature-Selection Approaches (10/17/25)**

I re-ran my **Denoising Autoencoder (DAE)** model and expanded the number of selected features from *k = 10* to *k = 15* and *k = 20* to increase overlap potential with other deep-learning methods. The training was performed on the breast_cancer(in).csv dataset, using Gaussian noise regularization and early stopping to prevent overfitting. After training, I ranked all 1,881 miRNAs using permutation-based importance and exported the top 15 and top 20 as separate files (dae_k15_features.csv, dae_k20_features.csv). These updated feature lists were shared with the team for integration into Anika's deep-learning overlap analysis.

k=15

```
miRNA
hsa-mir-323a
hsa-mir-323b
hsa-mir-431
hsa-mir-370
hsa-mir-485
hsa-mir-539
hsa-mir-329-1
hsa-mir-433
hsa-mir-432
hsa-mir-127
hsa-mir-382
hsa-mir-134
hsa-mir-136
hsa-mir-495
hsa-mir-493
```

k=20

```
miRNA
hsa-mir-323a
hsa-mir-323b
hsa-mir-431
hsa-mir-370
hsa-mir-485
hsa-mir-539
hsa-mir-329-1
hsa-mir-433
hsa-mir-432
hsa-mir-127
hsa-mir-382
hsa-mir-134
hsa-mir-136
hsa-mir-495
hsa-mir-493
hsa-mir-487b
hsa-mir-770
hsa-mir-668
hsa-mir-154
hsa-mir-1197
```

Task 9.1: Apply Logistic Regression Classification on miRNAs Selected by Feature Selection Approaches

(10/17/25)

I applied **Logistic Regression (LR)** classification using miRNAs selected by both my **Variance Threshold (VT)** and **DAE (k = 15, 20)** feature-selection methods. The pipeline included data standardization (StandardScaler), a 70/30 train-test split, and the LR model configured with balanced class weights and multinomial loss. Evaluation metrics included accuracy and macro-F1 score. The classifier achieved accuracies ranging from **0.33 to 0.37** and macro-F1 scores between **0.14 and 0.20** across the feature sets, with the DAE-derived features performing more consistently.

```
Used 10 features (ignored 0 missing).
              precision    recall  f1-score   support

           0       0.21      0.11      0.15        80
           1       0.00      0.00      0.00        52
           2       0.48      0.63      0.55       179
           3       0.00      0.00      0.00        48
           4       0.02      0.25      0.03         4

    accuracy                           0.34       363
   macro avg       0.14      0.20      0.14       363
weighted avg       0.28      0.34      0.30       363

Accuracy: 0.339 | Macro-F1: 0.144

=== dae_k15_features.csv ===
Used 15 features (ignored 0 missing).
              precision    recall  f1-score   support

           0       0.33      0.07      0.12        80
           1       0.17      0.12      0.14        52
           2       0.50      0.64      0.57       179
           3       0.19      0.10      0.13        48
           4       0.02      0.25      0.03         4

    accuracy                           0.37       363
   macro avg       0.24      0.24      0.20       363
weighted avg       0.37      0.37      0.34       363

Accuracy: 0.366 | Macro-F1: 0.199

=== dae_k20_features.csv ===
Used 20 features (ignored 0 missing).
              precision    recall  f1-score   support

           0       0.30      0.10      0.15        80
           1       0.20      0.13      0.16        52
           2       0.48      0.60      0.53       179
           3       0.09      0.06      0.07        48
           4       0.02      0.25      0.04         4

    accuracy                           0.35       363
   macro avg       0.22      0.23      0.19       363
weighted avg       0.34      0.35      0.33       363

Accuracy: 0.347 | Macro-F1: 0.192
```

	Feature file	#Used	#Missing	Accuracy	Macro-F1
0	vt_significant.csv	10	0	0.339	0.144
1	dae_k15_features.csv	15	0	0.366	0.199
2	dae_k20_features.csv	20	0	0.347	0.192

Task 10.1:Apply Stage-Wise Traditional Feature Selection Using Variance Threshold (10/25/25)

For this task, I will apply the Variance Threshold (VT) method individually for each cancer stage comparison (Normal vs. Stage I–IV) to identify the most informative miRNAs per stage. Using the same preprocessing pipeline as previous tasks (StandardScaler, train_test_split with 70/30 split), I will calculate feature variances and retain only the top 15 and top 20 features for each stage. These stage-specific feature lists will be saved as separate CSV files for comparison with the deep learning and clustering approaches. Performance metrics will serve only to verify that feature selection behaved as expected.

Task 10.2 : Apply Stage-Wise Deep Learning Feature Selection Using DAE (10/25/25)

For this task, I will extend my previous DAE workflow to a stage-wise setup, running separate autoencoders for each stage comparison (Normal vs. Stage I–IV). Each DAE will include Gaussian noise regularization, dropout layers, and early stopping to avoid overfitting. After training, I will compute feature importance scores using permutation error and select the top 15 and top 20 miRNAs for each stage. The results will be exported as CSV files (dae_stagewise_k15.csv, dae_stagewise_k20.csv) for later integration with other team methods during cross-comparison and classification tasks.

Task 10.3: Apply Stage-Wise Clustering as Feature Selection Using Spectral Clustering (10/25/25)

For this task, I will apply Spectral Clustering for each stage comparison (Normal vs. Stage I–IV) to identify groups of miRNAs that exhibit distinct expression patterns. Using the same preprocessing pipeline (scaling and data transposition) from previous tasks, I will construct similarity graphs for each stage and cluster the samples into two groups based on the eigenvector decomposition of their affinity matrix. Following Dr. Ali's guidance, the cluster showing the strongest separation or smallest membership will be considered the "important" cluster. From these clusters, I will extract the top 15 miRNAs per stage and save them as individual CSV files. These outputs will later be compared with the traditional (VT) and deep-learning (DAE) methods during the stage-wise analysis.